

GATE

ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 04



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Recap of Previous Lecture



Topic

Averages



Topics to be Covered



Topic-1

Percentages



AVERAGE



Sum of obs.

—————
No. of obs.

[MCQ]

$$A = \frac{\text{Sum}}{\text{No}}$$



#Q. What would be the average of: 1, 2, 3, 4, 5,49, 50.

$$\sum n = \frac{n(n+1)}{2}$$

$$\text{Sum} \rightarrow \cancel{1250} - \underline{1275}$$

$$\begin{aligned} x &= 1 + 2 + 3 + 4 + 5 + \dots + 49 + 50 \\ x &= 50 + 49 + 48 + \dots + 2 + 1 \end{aligned}$$

$$A \rightarrow \underline{\underline{\frac{n+1}{2}}}$$

$$\frac{51}{2} = \underline{25.5}$$

$$2x = 50 \times 51$$

$$x = \frac{50 \times 51}{2} = \underline{1275}$$

Average of Even & Odd

$$N \rightarrow \frac{n+1}{2}$$

$$E \rightarrow \underline{n+1}$$

$$O \rightarrow \underline{n}$$

Sum of Even & Odd



$$A = \frac{\text{Sum}}{\text{No.}}$$

$$\Rightarrow \underline{A} \times \text{No.} = \underline{\text{Sum}}$$

$$\textcircled{O} \Rightarrow n \times n = \textcircled{\begin{matrix} 2 \\ n \end{matrix}}$$

$$\textcircled{50 - 90}$$

51

89

70

$$\textcircled{E} \Rightarrow (n+1) \times n \\ = \textcircled{n(n+1)}$$

[MCQ]



#Q. What is the average of first five multiples of 12?

A 42

B 40

C 36

D 48

12, 24, 36, 48, 60 $\Rightarrow$ 36

[MCQ]



#Q. A class with 20 students has the average age as 14 years. When the teacher is included in the group, the average becomes 16 years. What would be the age of that teacher?

Logically

$$14 \times 20 = \underline{\underline{280}}$$

Formula

$$A = \frac{\text{Sum (S)}}{\text{No. (No.)}}$$

$$\Rightarrow A \times \text{No.} = \text{Sum}$$

$$14 \times 20 = \underline{\underline{280}}$$

$$16 \times 21 = \underline{\underline{336}}$$

$$\underline{\underline{56}}$$

[MCQ]



#Q. A team of 10 employees has the average age as 20 years. If team leader is excluded from the group the average of remaining 9 employees decreases by two months. Find the age of the team leader. ?

21.5 yrs
OR

21 yrs 6 months

[MCQ]

$$A = \frac{\text{Sum}}{\text{No.}}$$



#Q. In a school with 15 teachers, the average monthly salary is ₹4500. When three teachers left the school, the average monthly salary decreased by ₹600. Find the average monthly salary of three teachers who left the school.

$$4500 + 2400 = \underline{\underline{6900}}$$

$$\frac{20,700}{3}$$

$$A \times \text{No.} = \text{Sum}$$

$$4500 \times 15 = 67,500$$

$$3900 \times 12 = 46,800$$

$$\underline{\underline{20700}}$$

$$= \underline{\underline{6900}}$$

[MCQ]



#Q. The average score of a class of 40 students is 52. What will be the average score of the rest of the students if the average score of 10 of the students is

61.

A

47

B

49

C

50

D

48



$$52 - 3 = 49$$

[MCQ]



#Q. The average weight of a school of 40 teachers is 80 kg. If, however, the weight of the principle be included, the average decreases by 1 kg. What is the weight of the principal?

A

49

B

109

C

39

D

29

$$80 - 41 = \underline{\underline{39}}$$

[MCQ]



#Q. The average age of Abhijeet and Daya is 20 years. Their average age 5 years hence will be

Assignment

- A** 20
- B** 30
- C** 25
- D** 22

[MCQ]



#Q. The average monthly salary of 20 employees is ₹1500. If the manager's salary is added the average becomes ₹1600. The manager's salary is

Assignment

- A** ₹ 3500
- B** ₹ 3600
- C** ₹ 3800
- D** ₹ 3900

[MCQ]



#Q. Three years ago, the average age of a family of 5 members was 17 years. A baby having been born, the average of the family is the same today. What is the age of the baby?

Assignment

- A** 6 months
- B** 9 months
- C** 1 year
- D** 2 years

[MCQ]



#Q. 12 years ago, the average age of a husband and his wife was 20yrs. The average age is same today, they having two children. What is the present age of the youngest child if children differ in age by 2yrs?

Assignment

A

8

B

6

C

7

D

9

[MCQ]



#Q. The average of 5 consecutive integers starting with x is y . What is the average of 6 consecutive numbers starting with $(x+2)$?

Assignment

- A** $y + 3$
- B** $\frac{2y + 9}{2}$
- C** $y + 2$
- D** $\frac{2y + 5}{2}$

PERCENTAGE

Every 100

What?

(Use) Why?

$$\frac{793}{800}$$

$$\frac{585}{600}$$

15^m May, 2004

2000 → 0

J-3

F-0

M-3

A-2

M-1

Tuesday

~~Wednesday~~
~~Tuesday~~

1st March, 2000

→ 3 odd days
 $5 + 5 = \underline{10}$

1999 → 5

J - 3
 F - 1
 M - 1

 5

1900

↓
 1

99 → 4
 L-4 N-4
 24 75
 48 75

$6 + 5 = 11$

04
 x24
 96

PERCENTAGE



$$120\% = 1.2 = \frac{6}{5} \quad \underline{\underline{0.6}}$$

$$65\% \quad \checkmark$$
$$\left(66\frac{2}{3}\right)\% = \frac{2}{3} = \underline{\underline{0.6}}$$
$$70\% \quad \checkmark$$

$$30\% \quad \checkmark$$
$$\left(33\frac{1}{3}\right)\% = \frac{1}{3} = \underline{\underline{0.3}}$$

$$5\% = \frac{1}{20} = 0.05$$

$$30\% = \frac{3}{10} = \underline{\underline{0.3}}$$

$$10\% = \frac{1}{10} = 0.1$$

$$15\% = \frac{3}{20} = 0.15$$

$$20\% = \frac{1}{5} = \underline{\underline{0.2}}$$

$$100\% = \checkmark = \underline{\underline{1}}$$

$$25\% = \frac{1}{4} = 0.25$$

PERCENTAGE

Successive ✓



Next
Session

$$50\% \downarrow + 50\% \downarrow \rightarrow 75\% \downarrow$$

$$50\% \downarrow + 20\% \downarrow \rightarrow 60\% \downarrow$$

$$50\% \uparrow + 50\% \downarrow \rightarrow 25\% \downarrow$$

$$10\% \uparrow + 10\% \uparrow + 10\% \uparrow \rightarrow \underline{\underline{33.1\% \uparrow}}$$



2 mins Summary



Topics

✓
Averages & Percentages *✓*



THANK - YOU